Juhyun 'Simon' Park

https://parksimon0808.github.io/ juhyunp at princeton.edu

Education

Princeton University *Princeton, NJ, USA*
Ph.D. Student in Computer Science (Adviser: Sanjeev Arora) *2025 - Current*
M.S.E. in Computer Science (Adviser: Sanjeev Arora) *2023 - 2025*
A.B. in Mathematics, Cum Laude *2017 - 2023*

Interests

LLMs, Reasoning, AI for Math

Publication

Ortega, **Park**, Khaitan, Fowl, Kontorovich, Arora, "Do Theorem Provers Need Theorem-Proving Agents? mini-Lean-agent: A Coding-Agent Baseline for Lean," NeurIPS 2026 Workshop. [link]

Chung, Cai, Li, Yin, Agarwal, **Park**, Porto, Ri, Yang, Tang, Dang, Lin, Wang, Chen, Jin, Fowl, Arora, "Goedel-Architect: Streamlining Formal Theorem Proving with Blueprint Generation and Refinement," Preprint. [link]

Park*, Kaur*, and Arora, "How Does RL Post-training Induce Skill Composition? A Case Study on Countdown," NeurIPS 2026. [link]

Park*, Panigrahi*, Cheng*, Yu, Goyal, and Arora, "Generalizing from SIMPLE to HARD Visual Reasoning: Can We Mitigate Modality Imbalance in VLMs?," ICML 2025. [link]

Kaur*, **Park***, Arora, and Goyal, "Instruct-SkillMix: A Powerful Pipeline for LLM Instruction Tuning," ICLR 2025. [link]

Shah, Yu, Lyu, **Park**, Yu, He, Ke, Mozer, Bengio, Arora, and Goyal, "AI-Assisted Generation of Difficult Math Questions," NeurIPS 2024 Workshop. [link]

Professional Service

Organizer Princeton Language Intelligence Seminar Lunch Series [link]
Lecturer Guest Lectures on LLMs at Princeton Research Computing [1, 2, 3]
Reviewer NeurIPS (26), ICML (26), ICLR (26, 27)

Awards

Kwanjeong Educational Foundation Scholarship *Sep 2025 - Current*
Kwanjeong Educational Foundation

Gordon Wu Fellowship *Sep 2025 - Current*
Princeton University, Top Incoming Ph.D. Students in Engineering

Outstanding Student Teaching Award *May 2023*
Princeton University Department of Computer Science

Shapiro Award for Academic Excellence *Sep 2019*
Princeton University, Top 3% of Class

Teaching Experience

Natural Language Processing Graduate TA *Spring 2025*
Introduction to Machine Learning Head TA *Spring 2024, Fall 2023*
Natural Language Processing Undergraduate TA *Spring 2023*
Introduction to Machine Learning Undergraduate TA *Fall 2022, Spring 2023*

Skills

Programming Languages: Fluent in Python, Java / Familiar with C, R, SQL
Natural Languages: Native in Korean / Fluent in English, Mandarin Chinese